CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – V

MIX DESIGN & SPECIAL CONCRETES

Syllabus covered in this unit

Ready mixed concrete – fibre reinforced concrete (types of fibres, factors) – high performance concrete – self-consolidating concrete – self-healing concrete – factors in choice of mix proportions – quality control – statistical methods – acceptance criteria – ACI method & IS code method of mix design

1. READY MIXED CONCRETE (RMC)

- **RMC** = concrete batched at a **central plant** under controlled conditions and **delivered in plastic state** to the site by transit mixers (IS 4926 : 2003).
- Types : **central-mixed, transit-mixed, shrink-mixed** (see Unit II).
- Plant : aggregate bins & silos (cement, fly ash), weigh batchers (accuracy ± 2 %), water & admixture dosing, control room (computerised), truck mixers (4–9 m^3, 2–3 rpm agitation, 6–18 rpm mixing), pumps/boom pumps.
- **Advantages** : consistent quality (SD ≈ 3.5 N/mm^2), faster construction, less labour & storage at site, cleaner, minimal wastage, use of mineral & chemical admixtures, high-grade concrete (M30–M80).
- **Disadvantages** : traffic delay, slump loss (retarders, SP), not economical for small quantities, plant investment, need for pump.
- **Quality control** : fresh tests (slump, temperature, air) at delivery; cubes 28 d; delivery docket details (grade, slump, time of batching, quantities).
- Specification : **designed mix (producer responsible for strength)** vs **prescribed mix** (purchaser specifies proportions).

2. FIBRE REINFORCED CONCRETE (FRC)

Definition ★

FRC = concrete made of cement, aggregates (≤ 10–20 mm) and **discrete, randomly oriented short fibres** (0.1–2.0 % by volume) to improve **tensile strength, toughness, crack control and impact resistance.**

2.1 Types of fibres

Steel fibre shapes (increasing anchorage →)



Straight Crimped Hooked-end Deformed Twisted

Aspect ratio l/d = 50 – 100 ; length 20–60 mm ; dia 0.3–1 mm ; V_f 0.25 – 2 %

Fibre	Properties	Remarks / use
Steel (SFRC)	Dia 0.3–1 mm, length 20–60 mm, E = 200 GPa, f_t = 0.8–2 GPa ; V_f 0.25–2 % ; hooked-end, crimped, deformed	Pavements, industrial floors, tunnel shotcrete, precast, airfield; rust at crack surface only
Glass (GFRC, AR-glass)	Alkali-resistant (Zr-coating), E = 70 GPa, f_t = 1.7–3.5 GPa ; 2–5 % by weight ; l = 12–25 mm	Facade panels, cladding, architectural, thin sheets ; ordinary E-glass attacked by alkalis
Polypropylene (PP)	Low modulus (3.5–8 GPa), f_t 0.4–0.7 GPa, hydrophobic, chemically inert ; V_f 0.1–0.5 % ; l = 12–19 mm	**Plastic shrinkage crack control**, fire spalling resistance, shotcrete
Carbon	E = 230–400 GPa, f_t = 2–3 GPa ; costly	High strength, cladding, blast
Natural / organic	Jute, coir, sisal, bamboo, hemp, cellulose ; low cost, biodegradable ; l ≈ 25–50 mm	Low-cost roofing sheets, rural housing ; durability issue (alkali degrades)
Asbestos (obsolete)	E = 160 GPa ; health hazard	No longer used
Nylon, polyester, basalt, aramid	Elastic, impact resistant ; basalt = high temp. resistance	Specific uses

2.2 Factors affecting properties of FRC

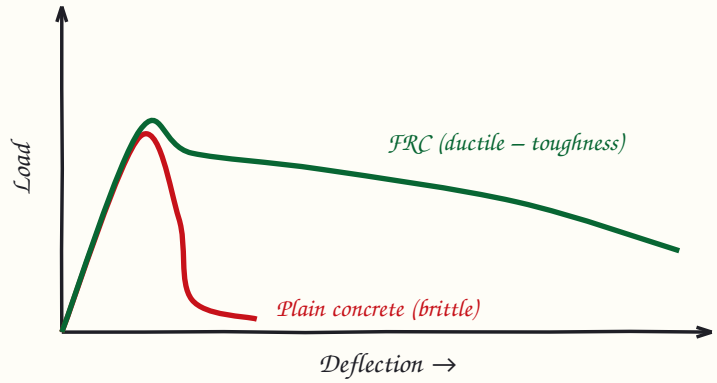
Key parameters ★

1. **Fibre type & material** (modulus, strength, bond) 2. **Aspect ratio (l/d)** 3. **Volume fraction V_f** 4. **Orientation & distribution** 5. **Matrix strength (w/c)** 6. **Aggregate size & content** 7. **Mixing, placing & compaction method** 8. **Bond (shape, surface)**

▶ **Aspect ratio** (l/d) : 50–100 ; > 100 → balling (poor mixing) ; < 30 → pull-out. Greater

aspect ratio improves strength & toughness but lowers workability.

- **Volume fraction** : up to 2 % for steel (> 2 % balling) – strength ↑ almost linearly ; effective at V_f > **critical volume**.
- **Orientation** : random (3D) efficiency factor ≈ 0.41 (2D 0.64, aligned 1.0) ; vibration orients fibres along flow ; thin section → 2D.
- **Max aggregate size ≤ 10–12 mm** (larger sizes obstruct fibre ; mortar-rich mix, cement ≥ 350–450 kg/m^3, fly ash, sand 50–70 % of aggregate).
- **Bond** : pull-out of smooth fibres dominates ; deformed / hooked ones fail by fibre fracture.
- **Mixing** : add fibres last, through vibrating screen / belt dosing, using SP ; avoid balling.



Fibres bridge cracks :
- ↑ flexural & tensile strength
- ↑↑ toughness & impact
- ↑ fatigue, shear, abrasion
- ↓ shrinkage cracks
- Compressive strength ≈ same
- Workability ↓

2.3 Properties and mechanics of FRC

- **Compressive strength** : unchanged / ↑ 0–25 % ; **flexural strength** ↑ 40–150 % (steel 1–2 %) ; **split tensile** ↑ 30–50 % ; **impact** ↑ 3–10× ; **toughness index** ↑ 10–40× ; **fatigue endurance limit** ↑ (0.65 → 0.9 f_r) ; **shrinkage** ↓ ; **durability** (freeze thaw) similar.

Fibre factor & Composite law

Fibre factor $\mathcal{F} = \beta\ (l/d)\ V_f$ (β = bond factor : 0.5 round, 0.75 crimped, 1.0 indented)

Composite $\sigma_{cu} = \sigma_m V_m + \eta\ \sigma_f V_f$

Critical fibre length $l_c = \sigma_f\ d\ /\ (2\ \tau_b)$ (τ_b = bond stress) – effective only when $l > l_c$; pull-out when $l < l_c$

Flexural strength (Swamy) : $\sigma_{cu} = 0.97\ \sigma_{cm}\ V_m + 3.41\ (\tau\ l\ /\ d)\ V_f$

- **Applications** : industrial floors / **pavements** & airport runways, **tunnel linings (shotcrete)**, canal lining, bridge decks, precast pipes, dams (repair), refractory, blast-resistant & seismic structures, thin shell, **overlays**.
- **Limitations** : workability, cost, balling, fibre corrosion at cracks (steel), nonuniform

distribution, fire (PP melts).

3. HIGH PERFORMANCE CONCRETE (HPC)

Definition (ACI) ★

HPC = concrete meeting **special combinations of performance & uniformity** (workability, strength, durability, volume stability, early strength) **that cannot be achieved routinely with conventional constituents & practices.** HSC (high-strength) ⊂ HPC. IS 456 : HSC = M65 & above.

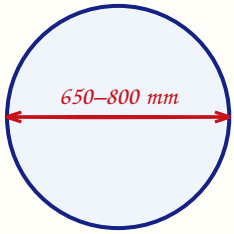
- **Characteristics** : strength 60 – 120 MPa ; very low w/b (**0.25 – 0.35**) ; high workability (slump > 200 mm with SP) ; high early strength ; high E (35–50 GPa) ; very low permeability (< 10^{-13} m/s) ; high durability (chloride, sulphate, carbonation) ; low creep & shrinkage ; **very brittle**.
- **Materials & mix** : Cement 400–550 kg/m^3 (OPC 53) ; **silica fume 5–10 %** ; fly ash / GGBS 15–40 % ; **superplasticizer (PCE) 1–2 %** ; crushed hard aggregate MSA 10–12.5 mm (strong, low absorption) ; w/b 0.25–0.35 ; VMA / SRA optional ; **cementitious total ≤ 600 kg/m^3**.
- **Mix-design approach** : (i) low w/b by SP (ii) pozzolan to fill voids & densify ITZ (iii) good aggregate (iv) optimum paste volume ; trial mixes.
- **Special care** : thorough mixing (long, 3–5 min), tight QC, **curing** is essential (early & wet 7–14 d ; internal curing by saturated lightweight aggregate / SAP for autogenous shrinkage), thermal control (high heat – cooling of mass).
- **Applications** : high-rise columns (Burj Khalifa 80 MPa), long span bridges, offshore platforms, nuclear containment, pavements, marine structures, precast / prestressed.
- **Advantages** : smaller sections ↑ usable area ; longer service life (> 100 yrs) ; lower life-cycle cost ; lower creep, shrinkage.

4. SELF-COMPACTING CONCRETE (SCC)

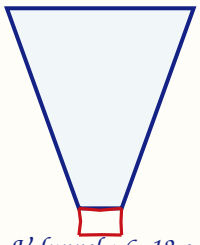
- SCC (Okamura, Japan, 1988) flows under its **own weight**, fills formwork & encapsulates reinforcement **without vibration**, while remaining homogeneous (EFNARC 2002 / IS 456 Annex /IS 10262 guidelines).
- **Key properties** : (1) **Filling ability** (flowability) (2) **Passing ability** (through congested

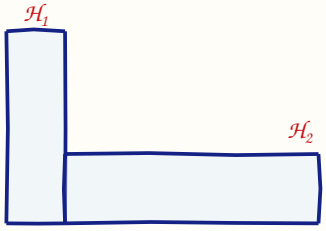
rebars) (3) **Segregation resistance** (stability).

- **Mix differences vs normal concrete** :
  - Higher **powder content** (cement + fly ash/GGBS/limestone filler/silica fume) 450–600 kg/m^3 ; **w/powder ratio 0.80–1.10 by volume** (w/p ≈ 0.28–0.40 by mass).
  - Lower **coarse aggregate** (28–35 % of concrete volume ; ≈ 750–900 kg/m^3) with MSA 12–20 mm ; sand 48–55 % of total aggregate.
  - **Superplasticizer (PCE)** (flow) + **VMA** (viscosity modifying agent – cohesion).
  - Water 150–200 kg/m^3.



Slump flow


V-funnel : 6–12 s



L-box : $H_2/H_1 \geq 0.8$


U-box : fill ≤ 30 mm

Test	Measures	Acceptance (EFNARC)
Slump flow (Abram's cone)	Filling ability, flow diameter ; T_{50} (time to 500 mm)	650 – 800 mm ; T_{50} = 2 – 5 s
V-funnel	Filling ability / viscosity	Flow time 6 – 12 s ; (V-funnel at 5 min ≤ +3 s : segregation)
L-box	Passing ability (blocking)	$H_2/H_1 \geq 0.8$
U-box	Passing ability	Height difference ≤ 30 mm
J-ring	Passing ability (with rebars)	Difference from slump flow ≤ 10 mm (≤ 50 mm acceptable)
Sieve segregation (5 mm sieve)	Segregation resistance	≤ 15 % (5–15 %), > 30 % unacceptable
Column segregation	Stability	≤ 15 %

- **Advantages** : no vibration (noise, labour), easy placing in congested sections, better surface finish & bond, faster construction, durable ; precast & RMC friendly.
- **Disadvantages** : costlier (powders, SP, VMA), need strict quality control, formwork pressure higher (full hydrostatic), sensitive to water variation.
- **Applications** : congested RC members, precast, tunnels, underwater, retrofitting, architectural, repair.

5. SELF-HEALING CONCRETE

- Concrete that **repairs its own cracks** automatically, restoring watertightness / durability (cracks < 0.1–0.3 mm for natural, up to 0.8 mm for engineered).

5.1 Autogenous healing (natural)

- Continued hydration of **unhydrated cement** (exposed in crack face) ; **carbonation** of $Ca(OH)_2$ → $CaCO_3$ crystal (dominant) ; swelling of C-S-H ; debris blocking. Needs water (wet–dry cycles) & CO_2, w/c < 0.4 (more unhydrated cement), fly ash/GGBS (late hydration) & crystalline admixtures.

5.2 Autonomous (engineered) healing

Technique	Mechanism	Remarks
Bacterial (bio-concrete)	Spores of *Bacillus pasteurii / pseudofirmus / subtilis (Sporosarcina)* + nutrient (calcium lactate) in clay pellets / capsules ; on crack & water ingress the bacteria wake & precipitate $CaCO_3$ (calcite)	TU Delft (Jonkers) ; seals 0.5–0.8 mm cracks in weeks ; viable > 200 yr (spores)
Microcapsules / polymer	Brittle capsules with epoxy / polyurethane / sodium silicate / MMA break at the crack & release healing agent	Quick, but one-time ; sizes 10–200 µm
Vascular network	Hollow glass tubes / fibres or 3D network supply healing agent from reservoir	Repeated healing ; complex, costly
Super-absorbent polymers (SAP)	Swell on water entry and block the crack; also internal curing	Sealing + extra hydration
Shape-memory polymers / ECC	SMP fibres contract when heated, closing cracks ; ECC (engineered cementitious composite) keeps crack widths < 60 µm so autogenous healing works	Research / niche
Crystalline admixture	Reactive chemicals precipitate crystals in the presence of water	Commercial (e.g. waterproofing)

- **Advantages** : longer service life, reduces maintenance cost, ↓ permeability, protects steel.
  Limitations : cost, limited crack width, viability of agents during mixing & high pH, strength reduction (capsules), standardisation lacking.

6. FACTORS IN THE CHOICE OF MIX PROPORTIONS

Mix design = determining the proportions of cement, water, FA, CA (and admixtures) to produce concrete of **required strength, workability, durability at minimum cost.**

1. **Compressive strength (target mean)** $f'_{ck} = f_{ck} + 1.65\ s$ (or $f_{ck} + X$ per IS 10262 : 2019).
2. **Workability** (placing, compaction, section thickness).
3. **Durability** – exposure condition → max w/c & min cement content (IS 456 Table 5).
4. **Type & grade of cement** (OPC 43/53, PPC), **aggregates** (MSA, shape, grading, zone, specific gravity).
5. **Maximum size of aggregate** (≤ ¼ min. member size) ; **admixtures** (SP, retarder) ; **mineral additions** (fly ash, GGBS).
6. **Quality control** – degree of supervision (standard deviation s) ; **economy** (cement content minimum).
7. Special : pumping, hot weather, mass concrete heat, shrinkage.

- **Methods** : arbitrary (1 : 2 : 4), fineness modulus, maximum density (Fuller), **ACI**, **IS 10262**, DOE (British), Road Note No. 4, Talbot, **trial mix**.

Exposure (IS 456 Table 5)	Max w/c (RCC)	Min cement (kg/m^3) & grade
Mild	0.55	300 ; M20
Moderate	0.50	300 ; M25
Severe	0.45	320 ; M30
Very severe	0.45	340 ; M35
Extreme	0.40	360 ; M40

7. QUALITY CONTROL OF CONCRETE

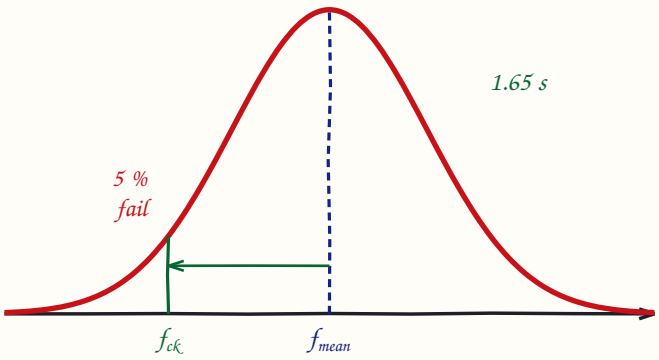
- **Quality control (QC)** = systematic procedures to ensure concrete **consistently meets specification** — covers **materials, mix proportions, batching, mixing, transport, placing, compaction, curing, testing, records.**
- **Essential points** : testing of cement (IS 4031), aggregates (IS 2386), water, steel ; calibration of batching plants ; periodic checks of **moisture content, bulking** ; slump & temperature at site ; **cube testing** (sampling frequency IS 456 cl. 15) ; maintaining **records** & control charts ; trained manpower ; supervision.
- **Variation causes** : materials (cement, aggregate grading), batching errors, mixing/handling,

testing errors (sampling, casting, curing, machine), site conditions.

Degree of control	Std. deviation s (N/mm^2)	Characteristics (ACI 214 – overall)
Excellent	< 2.8	Laboratory quality / RMC
Very good	2.8 – 3.5	Good plants
Good	3.5 – 4.2	Site with supervision
Fair	4.2 – 4.9	Average
Poor	> 4.9	Little control

- **Cost–quality** : high QC → lower s → lower target mean strength → **saving of cement** (cost).

8. STATISTICAL METHODS & ACCEPTANCE CRITERIA



Normal distribution of strength
- $f_{ck} = f_{mean} - 1.65\ s$ (5 % defectives)
- s = standard deviation
- $Cv = (s / f_{mean}) \times 100$ %
- f'_{ck} (target) = $f_{ck} + 1.65\ s$

- **Mean** $\bar{x} = \Sigma x / n$; **Standard deviation** $s = \sqrt{[\Sigma (x - \bar{x})^2 / (n - 1)]}$; **Coefficient of variation** $Cv = s / \bar{x} \times 100$ % (good < 10–15 %) ; **Range** $R \approx 2.5–3\ s$ for small samples.
- **Normal (Gaussian) distribution** of strength test results (symmetrical bell curve).

  Characteristic strength (95 % probability) $f_{ck} = f_m - 1.65\ s$; target mean strength $f'_{ck} = f_{ck} + 1.65\ s$.

Grade	s (IS 456 Table 8 / IS 10262 Table 2)	X (IS 10262 : 2019)
M10, M15	3.5	5.0
M20, M25	4.0	5.5
M30 – M50	5.0	6.5
M55 and above	6.0	8.0

- Target strength is the **higher** of ($f_{ck} + 1.65\ s$) and ($f_{ck} + X$) (IS 10262 : 2019). If s is from ≥ 30 test results, use actual s (IS 456 cl. 9.2.4).

Example

M25, $s = 4.0 \rightarrow f'_{ck} = 25 + 1.65 \times 4 = 31.6$; $25 + X = 25 + 5.5 = 30.5 \rightarrow$ target mean = **31.6 N/mm²**

Test data : 28.0, 31.5, 29.5, 33.0, 30.0 → mean 30.4 ; $s = \sqrt{[(2.4^2 + 1.1^2 + 0.9^2 + 2.6^2 + 0.4^2)/4]} = \sqrt{(14.7/4)} =$ **1.92** ; Cv = 6.3 %

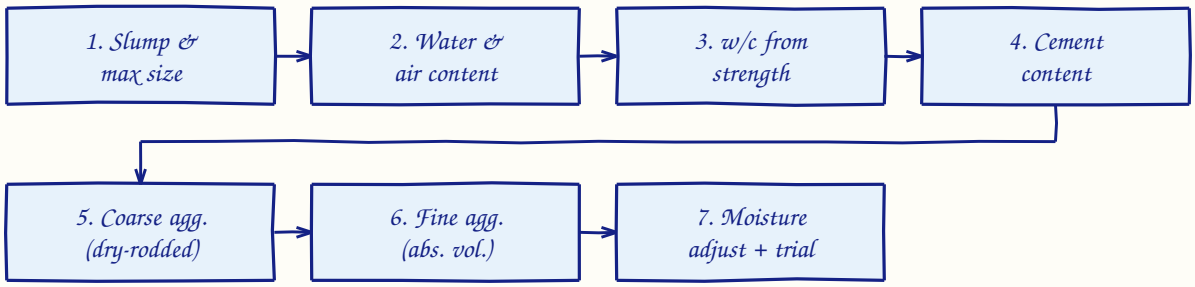
Acceptance criteria (IS 456 : 2000 cl. 16, Table 11) ★

- **Mean of any group of 4 consecutive test results** ≥ $f_{ck} + 0.825 \times s$ (established SD, rounded to nearest 0.5) or $f_{ck} + 3$ N/mm² (M20 and above), whichever is greater (for M15 and below : $f_{ck} + 0.825\ s$ or $f_{ck} + 3$, with no further rule).
- **Any individual test result** ≥ $f_{ck} - 3$ N/mm² (for M15 and below : ≥ $f_{ck} - 3$, i.e. 85 % f_{ck} for M10–M15 in older practice).
- Concrete is deemed to **comply** if both met ; else non-conforming → core / NDT / load test.

▶ **Control charts** ($\bar{x}$ & R charts) – upper/lower control limits ± 3 s help detect trends early. **Cumulative-sum (CUSUM)** also used. **Moving average of 7**.

9. ACI METHOD OF MIX DESIGN (ACI 211.1)

▶ Based on **empirical tables** & **absolute volume** method ; mix ratios per 1 m³ ; suits normal-weight concrete.



ACI 211.1 absolute-volume method – step flow

▶ **Step 1 : Slump** from Table (e.g. columns 25–75 mm, beams 25–100 mm, pavements 25–75 mm) ; **Step 2 : Max aggregate size**.

▶ **Step 3 : Mixing water** & air from tables (non-air-entrained) :

Slump (mm)	9.5 mm	19 mm	37.5 mm
25 – 50	207 kg/m³	190	166
75 – 100	228	205	181
150 – 175	243	216	190

Slump (mm)	9.5 mm	19 mm	37.5 mm
Entrapped air %	3.0	2.0	1.0

- **Step 4 : w/c** from strength (non-AE, 28-day mean) ; 45 MPa → 0.38 ; 40 → 0.42 ; 35 → 0.47 ; 30 → 0.54 ; 25 → 0.61 ; 20 → 0.69 ; 15 → 0.79. Also durability limits.
- **Step 5 : Cement content** = water / (w/c) (check ≥ minimum for durability).
- **Step 6 : Coarse aggregate** : bulk volume of dry-rodded CA per unit volume of concrete b/b_o from table (function of MSA & **FM of sand**), × dry-rodded unit weight.

Max size (mm)	FM = 2.4	2.6	2.8	3.0
9.5	0.50	0.48	0.46	0.44
19	0.66	0.64	0.62	0.60
25	0.71	0.69	0.67	0.65
37.5	0.75	0.73	0.71	0.69

- **Step 7 : Fine aggregate** = 1 m^3 – (water + cement + CA + air) volumes (absolute volume), × specific gravity. (Alternative : weight method, from estimated density of fresh concrete 2355 kg/m^3 for 19 mm).
- **Step 8 : Adjust for moisture & absorption** ; **Step 9 : trial batch**, check slump, strength, adjust.

ACI worked example

Data : target 30 MPa, slump 75–100 mm, MSA 19 mm, FM_{sand} = 2.6, G_c = 3.15, G_{FA} =

Water = **205 kg** ; air 2 % ; w/c (30 MPa) = **0.54** → cement = 205/0.54 = **380 kg**

CA = 0.64 × 1600 = **1024 kg**

Abs. volumes : water 0.205 + cement 380/3150 = 0.1206 + CA 1024/2700 = 0.3793 + air 0.020 = 0.7249 m^3 → sand volume = 0.2751 m^3 → **FA = 0.2751 × 2650 = 729 kg**

Mix per m^3 : Cement 380 : FA 729 : CA 1024 : Water 205 → **1 : 1.92 : 2.69, w/c 0.54**

10. IS CODE METHOD (IS 10262 : 2019)

- Applies IS 456 requirements ; for ordinary & standard concrete (M10–M70) with 28-day f_{ck} ; uses **w/c curves, water table & CA volume table**.

Steps ★

1. **Stipulations** : grade, MSA, cement type, exposure, workability, SG of materials, zone of sand, admixture.
2. **Target mean strength** : $f'_{ck} = f_{ck} + 1.65\ s$ or $f_{ck} + X$ (higher).
3. **w/c ratio** : from curve (cement strength vs 28-d strength) ; adopt lesser of this and IS 456 Table 5 limit.
4. **Water content** (Table 4 : 10 mm → 208, 20 mm → 186, 40 mm → 165 kg/m^3 for 25–50 mm slump, angular, Zone II) ; **+3 % per 25 mm slump above 50 mm** ; reduce 10 kg for rounded ; **SP reduces ≥ 20 %**.
5. **Cement content** = water / (w/c) ; check min (Table 5) & max 450 kg/m^3 (OPC) / up to 540 with SCM.
6. **CA volume fraction** (Table 5) : per unit volume of total aggregate for MSA & sand zone (at w/c 0.5) ; adjust ± 0.01 for each ∓ 0.05 change in w/c.
7. **Mix calculations** by absolute volume ; total aggregate volume = 1 − (cement + water + admixture volumes) ; CA & FA masses from fractions.
8. **Trial mixes** (3 trials) & adjust ; correct for moisture in aggregates.

Vol. of CA per unit vol. of total aggregate (w/c 0.5)	Zone IV	Zone III	Zone II	Zone I
10 mm	0.50	0.48	0.46	0.44
20 mm	0.66	0.64	0.62	0.60
40 mm	0.75	0.73	0.71	0.69

IS 10262 worked example (M25, moderate exposure)

Data : M25, OPC 43 (G = 3.15), 20 mm angular CA (G = 2.74), Zone II sand (G = 2.65),

Target = 25 + 1.65×4 = **31.6 MPa** ; w/c adopted = **0.46** (≤ 0.50 limit)

Water = 186 × 1.06 (50→100 mm : +6 %) = **197 kg** ; Cement = 197/0.46 = **428 kg** (≥ 300 ✓ ; ≤ 450 ✓)

CA fraction = 0.62 + (0.50 − 0.46)/0.05 × 0.01 = 0.628 ; FA = 0.372

Total aggregate vol = 1 − 428/3150 − 0.197 = 0.667 m^3

CA = 0.667 × 0.628 × 2.74 × 1000 = **1148 kg** ; FA = 0.667 × 0.372 × 2.65 × 1000 = **658 kg**

Mix : Cement 428 : FA 658 : CA 1148 : Water 197 = **1 : 1.54 : 2.68, w/c 0.46**

ACI vs IS 10262 – comparison

Feature	ACI 211.1	IS 10262 : 2019
Origin	USA, empirical tables	India, adapted from DOE / ACI, with IS 456 limits
Strength basis	28-day cylinder f'_c ($f'_{cr} = f'_c + 1.34\,s$)	28-day cube f_{ck} ; target = $f_{ck} + 1.65\,s$
Water	From slump & MSA table, air entrainment	Table 4 (+3 % / 25 mm slump) ; SP reduces
CA content	Dry-rodded volume (b/b_o) by FM of sand	CA fraction of total aggregate by zone
FA	Absolute volume / weight by difference	Absolute volume by difference
Durability	Separate exposure categories	IS 456 Table 5 limits

Quick revision – Unit V ★

- RMC : central, transit, shrink-mixed (IS 4926) ; FRC : steel/glass/PP, l/d 50–100, $V_f \le 2$ % ; HPC : w/b 0.25–0.35, SF + SP ; SCC : powder 450–600, slump flow 650–800 mm, V-funnel 6–12 s, L-box ≥ 0.8 ; self-healing : bacteria $CaCO_3$, capsules, vascular.
- $f_{ck} = f_m - 1.65\,s$; target = $f_{ck} + 1.65\,s$; acceptance : mean of 4 ≥ f_{ck}+0.825s (or +3), individual ≥ f_{ck}–3.
- ACI : water → w/c → cement → CA (dry-rodded) → FA (abs. vol.) ; IS 10262 : target → w/c → water (186 for 20 mm) → cement → CA fraction → masses.